
Personal Information

- **First name:** Isabel María
- **Last name:** Moreno Cuadrado
- **Date of birth:** 1 June 2001
- **Place of origin:** Granada, Spain

Education

- 2025–2029 **PhD in Advanced Mathematics**,
(expected) *Institute of Mathematical Sciences (ICMAT), CSIC*, Madrid, Spain.
 - Research area: **Quantum Information Theory**.
 - Position: **CSIC Severo Ochoa Programme Researcher**.
- 2024–2025 **MSc in Advanced Mathematics**,
Complutense University of Madrid, Madrid, Spain.
 - Research area: **Quantum Information Theory**.
- 2019–2024 **BSc in Mathematics**,
University of Granada, Granada, Spain.
 - Particular interest in Mathematical Analysis and Applied Mathematics.
 - *Honours* in the Bachelor's Thesis.
- 2019–2024 **BSc in Computer Science**,
University of Granada, Granada, Spain.
 - Specialisation in Artificial Intelligence.
 - *Honours* in the Bachelor's Thesis.
- 2017–2019 **High School Diploma**, *Regina Mundi School*, Granada, Spain.

Research Experience

- **Introduction to Research Programme (PI3).**
June 2025 – September 2025, IMUS, Seville, Spain.
Supervisor: Prof. Julián Eduardo Haddad
 - Fourier transform, distributions, and convex geometry: from the Busemann–Petty problem to modern functional analysis.
- **IAE INTRO ICU Research Contract.**
September 2024 – July 2025, CSIC, Spain.
Quantum Information Mathematics: foundations and applications, ICMAT.
Supervisors: Prof. David Pérez and Carlos Palazuelos
 - Bohnenblust–Hille inequality and applications in quantum learning theory.
- **Severo Ochoa Programme (INTRO-SO).**
June 2024 – July 2024, ICMAT, Madrid, Spain.
Supervisor: Prof. Manuel Garzón Martínez
 - Distribution theory and applications to PDEs.
- **ADIA Lab International Summer School.**
July 2024, University of Granada in collaboration with Abu Dhabi University.
 - Summer school/conference on trustworthy, secure, and governable AI.
 - Second prize for the project “Learning from Small Data”.
- **Undergraduate Collaboration Grant.**
October 2023 – July 2024, Department of Mathematical Analysis, University of Granada.
Supervisor: Prof. Francisco Javier Meri de la Maza
 - Research on distribution theory and its application to neural networks.

- **Summer Research Programme.**

July 2023 – September 2023, Donostia International Physics Center (DIPC), San Sebastián, Spain.

Supervisor: Prof. Thomas Frederiksen

- Application of mathematical and physical models to develop an app to explore the electronic structure of nanographene.

Awards and Scholarships

- **Blas Cabrera Summer School Scholarship**

August 2025, Menéndez Pelayo International University.

Scholarship programme supporting research initiation, teaching, and scientific innovation for 80 top-achieving university students.

- **Blas Cabrera Summer School Scholarship**

August 2024, Menéndez Pelayo International University.

- **Banco Santander Scholarship.**

September 2023 – June 2024, University of Granada.

- **UGR Skills Acceleration Programme.**

September 2023 – May 2024, Incubadora de Talento, University of Granada.

Selective programme offering training and professional development opportunities to 20 top-ranked students across disciplines.

Projects

- **Master's Thesis (with Honours).**

September 2024 – June 2025, Complutense University of Madrid, Madrid, Spain.

- Study of the Bohnenblust–Hille inequality and its applications in learning theory and quantum information theory.
Supervisors: [Prof. David Pérez](#) and [Carlos Palazuelos](#).

- **Bachelor's Thesis (with Honours).**

September 2023 – June 2024, University of Granada, Granada, Spain.

- Study of Fourier analysis in $\mathbb{R}^n$. Supervisor: [Prof. Francisco Javier Meri de la Maza](#).
- Analysis of the Fast Fourier Transform (FFT) to accelerate the training of Convolutional Neural Networks.
Supervisor: [Prof. Jesús Giráldez Crú](#).

- **Parking Space Detection Project in Granada.**

September 2023 – January 2024, Computer Vision course, University of Granada, Granada, Spain.

- Final project for the Computer Vision course.
- GitHub repository: [granada-parking-segmentation](#).

- **Fraud Detection in Financial Transactions Project.**

March 2024 – May 2024, Machine Learning course, University of Granada, Granada, Spain.

- Final project for the Machine Learning course.

Courses and Academic Activities

- **13th Heidelberg Laureate Forum (HLF).**

13–18 September 2026, Heidelberg, Germany.

- Selected as a Young Researcher participant.

- **Research seminar at the Shanghai Institute for Mathematics and Interdisciplinary Sciences (SIMIS).**

2026, Shanghai, China.

- Talk titled “Towards Lower Bounds in Position-Based Cryptography”.

- **Quantum Position Verification Workshop.**

13–17 July 2026, Lorentz Center, Leiden, The Netherlands.

- Poster presentation titled “Towards Exponential Lower Bounds in Position-Based Cryptography”.

- **Research seminar at Leiden University.**

16 July 2026, Applied Quantum Algorithms group, Leiden Institute of Physics (LION), Leiden University, The Netherlands.

- Talk titled “Quantum Cryptography and the Local Theory of Banach Spaces: A Hidden Connection”.

- **7th Seefeld Workshop on Quantum Information.**

21–26 June 2026, Seefeld, Tyrol, Austria.

- Poster presentation titled “Towards Exponential Lower Bounds in Position-Based Cryptography”.

- **Quantum Information Journal Club.**

June 2026, Instituto de Física Teórica (IFT UAM–CSIC), Madrid, Spain.

- Led a discussion on “Exponential Quantum Advantage in Processing Massive Classical Data”.

- **PhD Students Seminar UCM.**

29 January 2026, Complutense University of Madrid, Madrid, Spain.

- Talk titled "*Quantum Cryptography and the Local Theory of Banach Spaces: A Hidden Connection*".
- o **Attendance at the 7th BYMAT Conference: Bringing Young Mathematicians Together.**
 - Talk titled "The Bohnenblust–Hille inequality and its applications in learning theory".
- o **Invited to the Young Researchers Workshop.**

22–24 September 2025, UCM, Madrid.

 - Presentation titled "Bohnenblust–Hille inequality and applications in learning theory".
- o **IAE Mathematics Summer School.**

June–July 2024 and June–July 2025, ICMAT, Madrid, Spain.

 - Summer school organised annually by ICMAT for undergraduate and master's students interested in mathematical research and its applications.
- o **National Meeting of Mathematics Students (ENEM).**

21–25 July 2025, Granada.

 - Poster presentation titled "*Bohnenblust–Hille inequalities and applications in learning theory*".
 - Talk titled "*What can pure mathematics tell us about how we learn?*".
 - Informational session on academic careers.
- o **Attendance at the conference "Quantum Technologies for Young Researchers".**

8–11 July 2025, Madrid.

 - Poster presentation titled "*Bohnenblust–Hille inequalities and applications in quantum information theory*".
- o **Meeting of the Mathematical Analysis and Applications Network.**

Cullera, 3–7 March 2025.

 - Conference attendance.
 - Research project and presentation: "*Almost disjoint families*".
- o **Attendance at the Biennial RSME Congress 2025.**

Bilbao, 13–17 January 2025.

 - Attendance at Angela Capel (University of Cambridge) plenary lecture titled "*Many-Body Quantum Systems*".
 - Attendance at the Quantum Information Theory parallel session.
 - Poster presentation titled "*Fast Fourier Transform*".
 - Talk titled "*Fourier analysis in computer science*".
- o **Instructor for an Artificial Intelligence Workshop.**

29 November, University of Granada, Granada, Spain.

 - Delivered a workshop on AI concepts and applications for undergraduate and master's students.
- o **Attendance at the 6th BYMAT Conference: Bringing Young Mathematicians Together.**
 - Attendance at the Quantum parallel session.
 - Poster presentation titled "Behind the algorithm: the historical paper that shaped modern signal processing".
 - Talk titled "Computer science through the frequency spectrum".
- o **Attendance at the National Meeting of Mathematics Students (ENEM).**
 - July 2022, University of Valencia.
 - July 2023, University of Badajoz.

Community Involvement

- o Class representative for five consecutive years.
- o Member of the academic committee of the Mathematics degree programme.
- o Speaker at science outreach events and school talks.
- o Active participation in academic and social volunteering initiatives.

Languages

Spanish	Native
English	Advanced
Chinese	Intermediate
French	Beginner